Cell-Type Composition Prediction from H&E Images using Multi-Scale CNNs and Structured Inter-Cell Refinement

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***Abstract*—Predicting spatially resolved cell-type composition from Hematoxylin-and-Eosin (H&E) stained tissue images is a challenging problem, mainly because the target cell abundances are not independent and often show strong relationships with each other. In this work, we explore two complementary approaches for this task. First, we use a multi-scale convolutional neural network to capture both local and contextual visual information from tissue patches. Second, we introduce a two-stage refinement framework, where the initial predictions are corrected using learned inter-cell dependencies. Our experiments show that this refinement step significantly improves performance, with ElasticNet-based correction increasing the base ResNet18 out-of-fold Spearman score from approximately 0.45 to 0.62, and further post-processing pushing it to around 0.64. We also analyze the learned dependency matrix to understand relationships between different cell types, and briefly discuss alternative approaches such as coordinate-based features and spatial GCNs that did not perform as well.**

I. Introduction

Recent work has explored the use of deep learning models for predicting spatial gene expression and cell-type composition from histology images [1], [2].

Predicting the spatial distribution of cell types from Hematoxylin-and-Eosin (H&E) stained tissue images is an important problem in computational pathology. Given an image of a tissue section, the goal is to estimate the relative abundance of different cell types at specific spatial locations. This task is challenging because the visual appearance of cells in H&E images is indirect and often ambiguous, making it difficult to identify cell types purely from local image features.

Most approaches treat this as a standard image-to-regression problem, where a convolutional neural network (CNN) is trained to predict all cell-type abundances directly from image patches. However, this formulation ignores an important aspect of the problem: cell types are not independent. In real tissue, certain cell types tend to co-occur, while others may be mutually exclusive, leading to structured relationships in the target space.

In this work, we explore two complementary approaches to address these challenges. First, we evaluate a multi-scale CNN model that incorporates both local and contextual information from tissue images, improving the quality of the initial predictions. Second, we propose a two-stage refinement framework, where the outputs of a base model are further processed using a linear model that captures inter-cell dependencies. This second stage enforces consistency across predicted cell abundances and significantly improves performance.

Our experiments show that while improvements in the base model help in better feature extraction, the largest gains come from modeling the relationships between cell types. These results suggest that cell-type prediction from H&E images should be viewed not only as a vision problem, but also as a structured prediction task.

II. Methods

A. Base Models

We train convolutional neural networks (CNNs) to predict the abundance of 35 cell types from H&E image patches. As a baseline, we use a ResNet18 architecture that takes a fixed-size patch centered at each spatial location and directly outputs a 35-dimensional prediction vector. The model is trained using an L1 loss, which is well-suited for regression tasks involving continuous abundance values.

To improve feature extraction, we also evaluate a multi-scale CNN architecture that incorporates information at different spatial resolutions. This model processes three inputs: a central patch, a set of surrounding sub-patches arranged in a grid, and a larger context tile capturing a broader tissue region. Each input is passed through a separate convolutional encoder to extract features at different scales. The resulting feature representations are concatenated and fed into a fully connected decoder to produce the final predictions.

This design allows the model to capture both local cellular patterns and larger tissue-level context, which can be important for accurately estimating cell composition.

B. Two-Stage Refinement

While the base models predict all cell types simultaneously, they treat each output independently. However, in biological tissue, cell types exhibit structured relationships, including

co-occurrence and mutual exclusion patterns. Ignoring these relationships can lead to inconsistent predictions.

To address this, we introduce a two-stage refinement framework. In the first stage, a CNN model produces initial predictions for all cell types. In the second stage, these predictions are used as input features to a separate model that learns to correct them.

We model this refinement step using ElasticNet regression. For each target cell type, the model learns a linear combination of all predicted cell types:

$$\hat{y}_i = \sum_{j=1}^{35} w_{ij} x_j \tag{1}$$

where x_j are the predicted abundances from the first stage, and w_{ij} are learned coefficients. The ElasticNet formulation combines ℓ_1 and ℓ_2 regularization, encouraging sparsity while maintaining stability.

The refinement model can be written in matrix form as:

$$\hat{\mathbf{y}} = W\mathbf{x} \tag{2}$$

where $\mathbf{x} \in R^{35}$ are the predicted abundances from the base model and $W \in R^{35\times 35}$ is the learned coefficient matrix. The sparsity induced by ElasticNet ensures that only a subset of entries in W are significant, reflecting structured dependencies between cell types.

This formulation allows the model to use well-predicted (easy) cell types to improve predictions for more difficult ones, while also enforcing global consistency across all outputs.

To avoid data leakage, we train the second-stage model using out-of-fold (OOF) predictions from the first-stage model. This ensures that the refinement model is trained only on predictions from unseen data, preserving generalization. The refinement framework is applied on top of the ResNet18 baseline model. This allows us to isolate and evaluate the effect of structured prediction independently of improvements in feature extraction. While more complex base models, such as the multi-scale architecture, produce stronger initial predictions, integrating the refinement stage with these models requires careful handling of out-of-fold predictions to avoid data leakage. In this work, we focus on evaluating the refinement framework in a controlled setting, and leave its integration with more advanced architectures for future investigation.

C. Graph-Based Model

In addition to convolutional models, we explored a graph-based approach to incorporate spatial relationships between neighboring locations. We construct a graph where each node corresponds to a spatial point, and edges connect nearby nodes based on spatial proximity. A Graph Convolutional Network (GCN) is then used to propagate information across neighboring nodes and predict cell-type abundances.

The intuition behind this approach is that nearby regions in tissue may have similar cellular composition. However, in practice, this model did not outperform the CNN baseline, suggesting that spatial proximity alone does not capture the dominant structure in the data.

D. Experimental Setup

We evaluate all models using Leave-One-Slide-Out Cross-Validation (LOSOCV), where the model is trained on all slides except one and evaluated on the held-out slide. This ensures that the model is tested on completely unseen tissue samples and prevents slide-specific overfitting.

Performance is measured using the mean Spearman correlation across all spatial locations. All models are trained using standard optimization methods, with mean absolute error (L1 loss) as the primary training objective.

The ResNet18 models are trained using the Adam optimizer with a learning rate of 1×10^{-4} for 5–10 epochs depending on convergence. The ElasticNet model is trained with regularization parameters selected through validation, with α in the range $[10^{-3}, 10]$. Input patches are extracted around each spatial location at a fixed resolution.

E. Coordinate Alignment Study

During exploratory analysis, we observed that the provided spot coordinates were slightly misaligned with respect to the tissue images in some slides. To evaluate the impact of this misalignment, we manually applied small coordinate shifts for selected slides and trained models on both the original and shifted datasets.

We found that training on the shifted data led to faster convergence and more stable early training behavior. However, the final performance remained largely unchanged, indicating that coordinate alignment primarily affects optimization dynamics rather than predictive accuracy.

III. Results

A. Model Performance

Table I summarizes the performance of different models evaluated using Leave-One-Slide-Out Cross-Validation. The baseline ResNet18 model achieves a moderate Spearman correlation, which improves significantly after applying the second-stage refinement. The ElasticNet-based refinement leads to the largest performance gain, and additional quantile-based post-processing further improves the results.

TABLE I: Comparison of model performance (Spearman correlation)

Model	Spearman Score
ResNet18 (base)	0.45
ResNet18 + ElasticNet	0.62
ResNet18 + ElasticNet + Quantile	0.64
GCN Model	0.43
Multi-scale CNN	*0.6387*

B. Comparison with Public Benchmarks

To contextualize our results, we compare our performance with publicly available solutions from the Global AI Hackathon'25 competition. Several baseline and advanced approaches reported Spearman scores in the range of 0.50 to 0.68.

For instance, a ResNet-based model achieves a score of approximately 0.68, while graph-based approaches such as Graph Attention Networks (GATs) report lower performance around 0.50. Other visual and spatial deep learning pipelines achieve scores around 0.62–0.63.

Our best model, which combines a ResNet18 baseline with a structured refinement stage, achieves a Spearman correlation of approximately 0.64. This places our approach competitively within the range of strong baseline methods, while maintaining a simpler architecture and providing additional interpretability through the learned dependency structure.

Notably, our approach achieves comparable performance to stronger baseline models despite using a relatively simple backbone, highlighting the effectiveness of modeling inter-cell dependencies.

C. Correlation Analysis

As part of exploratory data analysis, we computed the correlation matrix between different cell types using the ground truth abundances. The results showed strong positive and negative correlations between several cell types, indicating that the presence of certain cells is closely related to others.

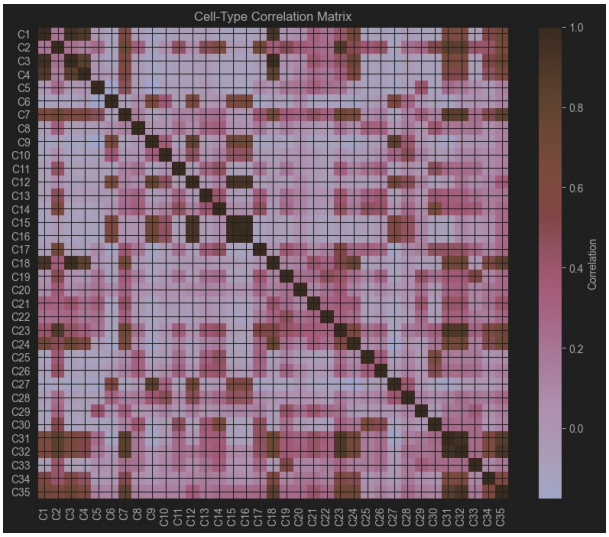

Fig. 1: Correlation matrix of ground truth cell-type abundances. Strong correlations suggest that cell types are not independent and exhibit structured relationships.

This observation motivated the use of a second-stage refinement model, where predicted cell abundances are corrected using learned relationships between cell types.

D. Inter-Cell Dependency Analysis

To better understand the effect of the second-stage refinement, we analyze the learned coefficients of the ElasticNet model. These coefficients represent how the prediction of each cell type depends on the predictions of other cell types.

The dependency matrix shows that only a subset of cell types significantly influence each target, indicating that the relationships are sparse. Additionally, some cell types consistently appear as strong contributors across multiple targets, suggesting the presence of structured clusters in the data.

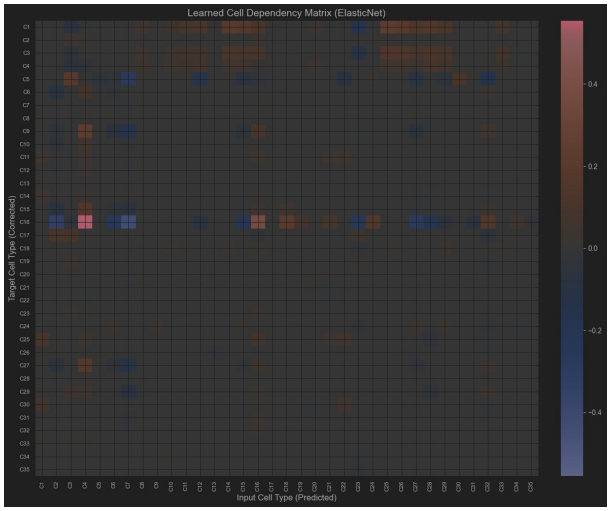

Fig. 2: Learned inter-cell dependency matrix. Rows correspond to target cell types, and columns correspond to influencing cell types. Positive and negative values indicate co-occurrence and mutual exclusion patterns respectively.

E. Key Observations

- The second-stage refinement provides the largest improvement in performance, indicating the importance of modeling inter-cell relationships.
- The learned dependency matrix reveals both positive and negative relationships, corresponding to co-occurrence and mutual exclusion of cell types.
- The dependency structure is sparse, with only a few cell types influencing each prediction.
- The GCN model does not improve performance, suggesting that spatial relationships alone are not sufficient for this task.

F. Comparison with Public Benchmarks

To contextualize our results, we compare our performance with publicly available solutions from the Global AI Hackathon'25 competition [3]. This competition focuses on predicting spatial cellular organization from histology images and provides a standardized evaluation benchmark :contentReference[oaicite:0]index=0.

Several approaches reported Spearman scores in the range of 0.50 to 0.68. For instance, ResNet-based models achieve scores around 0.68, while graph-based approaches such as Graph Attention Networks (GATs) perform significantly lower, around 0.50. Other deep learning pipelines incorporating spatial and multi-scale features achieve scores around 0.62–0.63.

Our best model achieves a Spearman correlation of approximately 0.64, placing it competitively within the range of strong baseline methods. Notably, this performance is achieved using a relatively simple backbone combined with a structured

refinement stage, highlighting the effectiveness of modeling inter-cell dependencies.

IV. Discussion and Conclusion

The results highlight that predicting cell-type composition from H&E images is not purely a visual recognition task, but also a structured prediction problem. While convolutional models are effective at extracting visual features, their outputs may not always respect the underlying relationships between cell types. The proposed two-stage framework addresses this limitation by refining predictions using learned inter-cell dependencies, leading to a significant improvement in performance.

The analysis of the dependency matrix further supports this observation. The learned coefficients reveal that only a subset of cell types strongly influence each target, indicating a sparse structure in the relationships. Additionally, both positive and negative dependencies are observed, which can be interpreted as co-occurrence and mutual exclusion patterns in the tissue. This suggests that enforcing consistency across predictions is important for accurate modeling of cell composition.

We also explored alternative approaches such as graph convolutional networks (GCNs) and coordinate-based features. However, these methods did not lead to improvements in performance. This indicates that spatial proximity alone may not capture the dominant structure in the data, and that modeling relationships in the prediction space is more effective for this task.

The multi-scale CNN model demonstrates that incorporating larger spatial context improves the quality of the base predictions by capturing broader tissue structure. On the other hand, the two-stage refinement model improves the consistency of predictions by leveraging inter-cell relationships. These two approaches address different aspects of the problem, and a unified model combining both remains an interesting direction for future work.

In conclusion, we show that incorporating structured refinement into the prediction pipeline significantly improves cell-type estimation from H&E images. Our results suggest that combining strong visual feature extraction with explicit modeling of inter-cell dependencies is key to achieving better performance on this task.

V. Figures and Tables

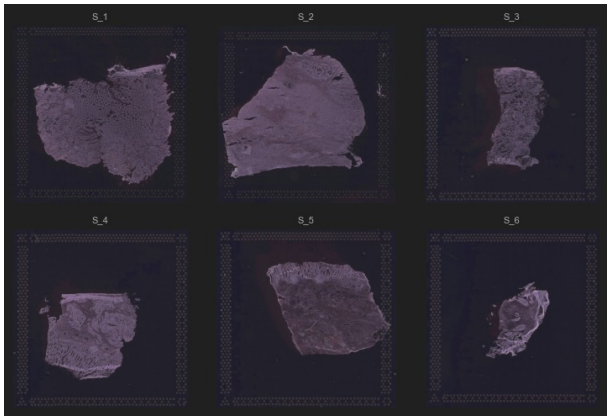

Fig. 3: H&E tissue slides showing the spatial structure of the data used for prediction.

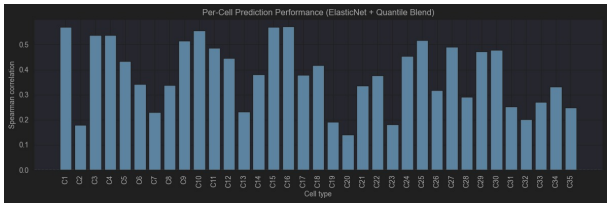

Fig. 4: Per-cell prediction performance. The variation in Spearman correlation across cell types indicates differing levels of prediction difficulty.

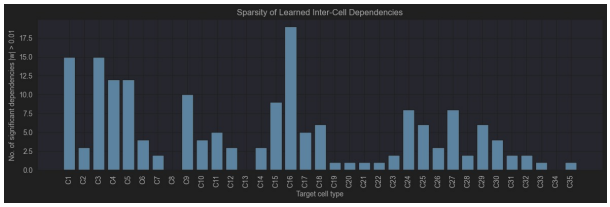

Fig. 5: Sparsity of learned inter-cell dependencies. Each cell type depends only on a limited number of other cells, supporting the use of sparse models.

References

[1] K. Nonchev, S. Dawo, K. Silina, H. Moch, S. Andani *et al.*, "DeepSpot: Leveraging Spatial Context for Enhanced Spatial Transcriptomics Prediction from H&E Images," *bioRxiv*, 2025. [Online]. Available: https://doi.org/10.1101/2025.02.09.25321567

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[3] Global AI Hackathon'25 by Elucidata, "Prediction of Spatial Cellular Organization from Histology Images," Kaggle Competition. [Online]. Available: https://www.kaggle.com/competitions/el-hackathon-2025/leaderboard